Preliminary rotor sizing according to API610 and Duncan & Hood diagram

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1. Purpose

This procedure defines the guidelines for preliminary rotor sizing according to API610 standard and Duncan & Hood diagram.

The procedure is applicable to the Nova Milanese and La Spezia plants.

2. Abbreviations

D shaft diameter (largest) at the impeller [mm]

D_1 diameter under the seal sleeve [mm]

D_2 diameter between bearings [mm]

K_t size factor

H corresponding total head [m]

I_{SF} shaft flexibility index

L_1 distance between the overhang centerline of the impeller and the line bearing [mm]

L, L_2 bearing span [mm]

n rotational speed [rpm]

Q flow at BEP [m³/h]

TIR total indicator reading or total indicated runout[1]

W rotor weight [N]

3. Basic knowledge

Sizing a shaft is an iterative process. In general, a minimum diameter is determined for strength, steps to enable assembly are added, deflection is checked, then the dynamic characteristics are predicted. The process is then repeated until an acceptable solution is realized.

Shafts require to have a certain level of strength to prevent failure due to fatigue. Said level of strength should consider the loads imposed during startup, normal operation, and any specified abnormal operation. Consequently, the endurance strength of the shaft material—considering the pumped liquid, if applicable—must exceed the combined stress at the shaft's critical point. The extent

[1] Difference between the maximum and minimum readings of a dial indicator or similar device, monitoring a face or cylindrical surface, during one complete revolution of the monitored surface.

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to which it must exceed this stress (i.e., the design factor) depends on several variables: the certainty of the applied loads, the condition of the liquid, the shaft detail at the critical point, and the material's condition.

Loading conditions should account for a 10% torsional fluctuation and, where applicable, include allowances for coupling misalignment.

Evaluating nominal torsional stress at the shaft's smallest diameter is a common approach. While conservative, it is not entirely reliable, as deflections and dynamic behavior are critical factors that must also be considered.

4. Overhung pumps

For overhung rotors, an acceptable design practice requires the following:

- The deflection, comprising both the static deflection and the one caused by imposed forces under the worst expected operating conditions, should be less than the available internal clearance.
- Deflection at the shaft seal should be less than 0.050 mm.
- First bending critical speed (dry basis) should be at least 20% above the running speed (except for high-speed pumps).

A simplified, and generally sufficiently accurate alternative approach, is to employ a factor that is proportional to the static deflection, which can be estimated based solely on the shaft geometry.

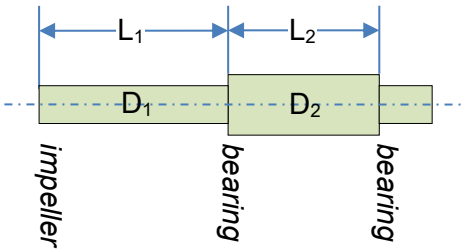

Figure 1: Overhung shaft (D_1<D_2).

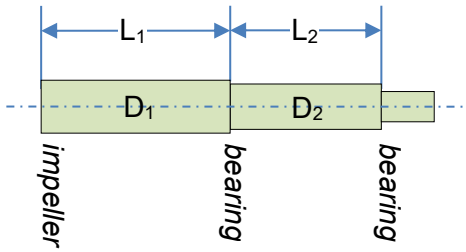

Figure 2: Overhung shaft (D_1>D_2).

L_1 = distance between overhang centerline of the impeller and the line bearing [mm]

L_2 = bearing span [mm]

D_1 = diameter under the seal sleeve [mm]

D_2 = diameter between bearings [mm]

For cantilever rotors of the form shown in *Figure 1* and *Figure 2*, the deflection is proportional to a parameter called shaft flexibility index I_{SF}:

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$$I_{SF} = {L_1^3}/{D_1^4} + {L_1 L_2^2}/{D_2^4}$$

In rotor designs typical of refinery pumps, $D_2 > D_1$ and $L_2 < L_1$, the second term accounts for approximatively 20% of the total value of I_{SF}. Therefore, the convention is to assess overhung pump shaft stiffness using a shortened expression:

$$I_{SF} \cong {L_1^3}/{D_1^4}$$

To provide general guidance on values of I_{SF}, it is necessary to relate them to the pump's size. The loads imposed on the shaft are directly related to the size of the impeller and, hence, to the pumps' flow rate, total head and rotational speed. This relationship allows the definition of a size factor K_T:

$$K_T = {QH}/{n}$$

$$Q = flow\ at\ BEP\ \left[{m^3}/{h}\right]$$

$$H = corresponding\ total\ head\ [m]$$

$$n = rotational\ speed\ [rpm]$$

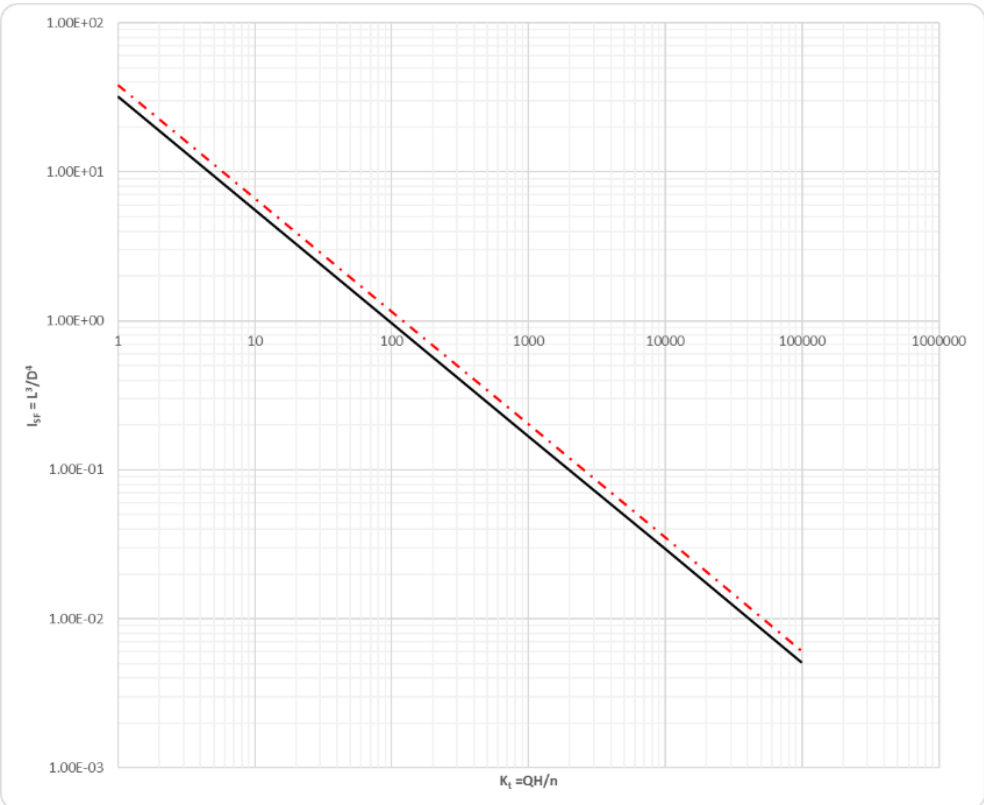

Figure 3: Shaft flexibility.

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For SI units, the following empirical formula covers refinery pumps whose rotors, at the limit of each frame, reportedly just meet the static deflection and rotor dynamics requirements of this standard based on rated speeds up to 3600 rpm. The liquid-end designs generally have double volutes for all pumps of 100 mm discharge and larger. In some cases, the designs may be limited to 3000 rpm.

$$I_{SF} = 32K_t^{-0.76}$$

Figure 3 and the previous equation can be used to make a first assessment of the rotor stiffness of a given overhung pump design or of similar designs for a given application. An overhung pump with an I_{SF} value bigger than 1.2 times the chart value or the equation value, requires to seek justification of the design from the pump's vendor.

5. Between bearing pumps

Between bearings rotors for one- and two-stage pumps are usually relatively short and therefore easily made to meet the simple requirements for cantilever rotors. Between bearings rotors for pumps having three or more stages cannot be designed to withstand any appreciable radial thrust. Twin volutes, diffusers, or staggered single volutes are employed to reduce radial thrust to negligible values. The design of the rotors themselves may follow two approaches. One approach, focused on lower cost and higher efficiency, advocates for low mechanical stiffness (slender shaft) and high hydraulic stiffness (Lomakin effect). The second approach claims that reliability is more important than initial cost and efficiency (the efficiency of a failed pump is zero) calls for high mechanical stiffness (large shaft) and low hydraulic stiffness.

In a more general sense, it is worth noting that this debate is peculiar to liquid handling turbomachines; those handling gases must rely on mechanical stiffness because the hydraulic stiffness is very low.

Slender shaft rotors are sized for torque, then scaled up to allow assembly and, in some cases, to meet empirical stiffness requirements. Their static deflection is greater than the available internal clearance, and the dry first bending critical speed is consequently significantly lower than the running speed.

As the pump accelerates, the Lomakin effect rapidly increases the hydraulic stiffness, so that at 100% speed, the rotor behaves as a classically "stiff" system (*Figure 4*). Most successful slender shaft pumps have opposed impeller rotors, a configuration in which the central bushing is particularly effective in generating the high Lomakin effect required for stable operation.

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Large-shaft rotors, on the other hand, are designed to ensure that static deflection remains below the available internal clearance. As a result, the dry first bending critical speed is significantly higher. Although the critical speed increases slightly as the pump accelerates, this change is minimal by design, and the rotor typically continues to behave as a classically "flexible" system (*Figure 4*).

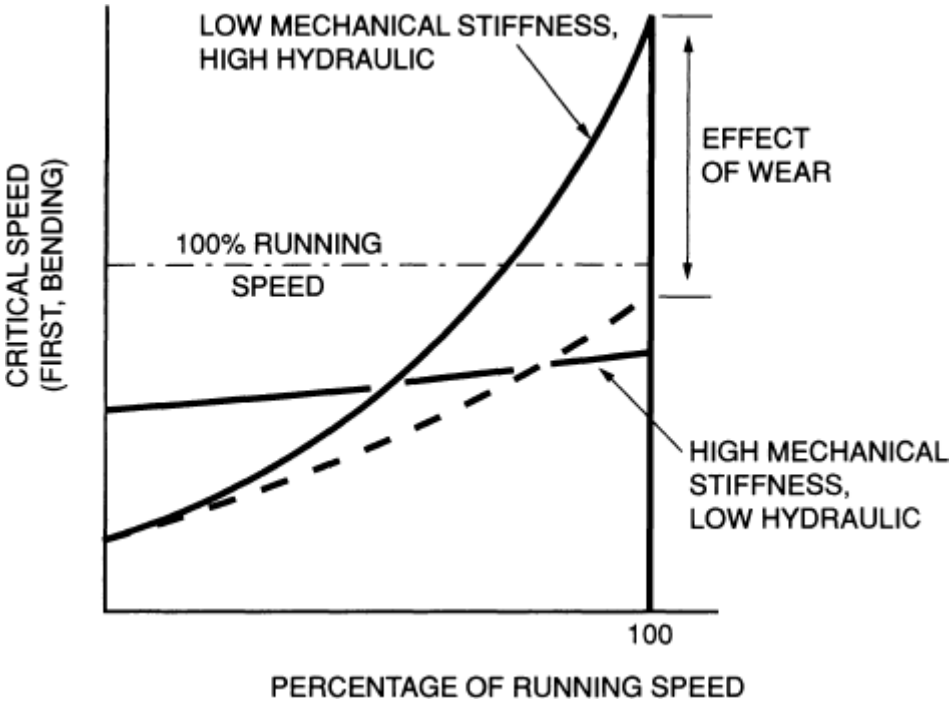

Figure 4: Critical speed vs running speed.

Given the Lomakin effect's dependence on pressure differential and running clearance condition, there are two cautionary points that should be considered. First, rubbing contact at the running clearances whenever the pump speed is significantly below design speed (e.g., during startup and shutdown), requires careful selection of materials to minimize damage. Even with appropriate materials, frequent start/stop operation may still lead to rapid wear. Second, the dynamic behavior of such rotors can be highly sensitive to the condition of the running clearances. In the worst-case scenario, wear may reduce the critical speed enough to coincide with the operating speed. If this occurs when damping is also significantly reduced due to worn clearances, the rotor may experience severe vibration, potentially causing further damage to the pump.

As with overhung rotors, a factor proportional to static deflection is frequently used to quickly evaluate between bearings rotors. The most common is the flexibility factor F_t:

$$F_t = \frac{L^4}{D^2}$$

$$L = bearing\ span\ [mm]$$

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$$D = largest\ shaft\ diameter\ at\ the\ impeller\ [mm]$$

This factor is based on the stiffness of the shaft alone and neglects the weight of the components mounted on it. Such an approach is justified on the grounds that the shaft provides most of the rotor's mechanical stiffness.

The guideline for F_t is defined in API610 and defines:

- Limits for allowable shaft runout, components fits and rotor runout (*Table 1*).
- A demarcation for rotor runout and balancing (*Table 2*).

Table 1: Shaft and Rotor Runout Requirements.

Flexibility Factor F_t^a $[mm^2]$	**Allowable Shaft Runout** TIR $[\mu m]$	**Component Fit on Shaft**	**Allowable Rotor Radial Runout** TIR^b $[\mu m]$
>1.9E9	40	Clearance	90
		Interference	60
≤1.9E9	25	Clearance	75
		Interference	50

[a] F_t is directly related to the static deflection of a simply supported shaft and is a good indicator of the runout attainable during manufacture and the quality of balance that can be achieved and maintained.

[b] Runout of impeller hubs, balancing drum, and sleeves.

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Table 2: Rotor Balance Requirements.

Component Fit on Shaft	Maximum continuous speed [rpm]	Flexibility Factor $F_t\ [mm^2]$	Rotor Balance Procedure(s)[b]	Rotor Balance Grade
Clearance	≤3800[a]	no limit	C	[c]
Interference	≤3800	no limit	C + B or D	G2.5[d]
	>3800	1.9E9	C + B or D	G0.67[d e]

[a] to allow for a 5% speed increase.

[b] see ISO 11342.

[c] balance correction during assembly is not feasible because clearance fit does not maintain corrected balance.

[d] approximately equal to the midpoint of the corresponding ISO balance quality grade.

[e] if rotors of higher flexibility are used at speeds above 3800 r/min, achieving and maintaining this balance level requires special attention to design, manufacturing, and maintenance.

It is important to highlight that neglecting the weight of the mounted components distorts the evaluation, particularly for slender shaft rotors whose mounted components usually weigh more than the shaft and therefore have a significant influence on static deflection and dry critical speeds.

A most appropriate rules is based on Duncan and Hood studies.

5.1. Duncan & Hood diagram

There is a factor proportional to the rotor's first dry bending critical speed for the evaluation of multistage horizontal pumps[2]:

$$K = \sqrt{\frac{WL^3}{D^4}}$$

$$W = rotor\ weight\ [N]$$

$$D = largest\ shaft\ diameter\ at\ the\ impeller\ [mm]$$

$$L = bearing\ span\ [mm]$$

[2] It is very important to highlight that stiffening/damping/mass effects of wear ring, pistons, etc… are not considered in this approach and only rotor stiffness and mass are used to define the shaft's size; this means that a selection is suitable to work with different operating fluids.

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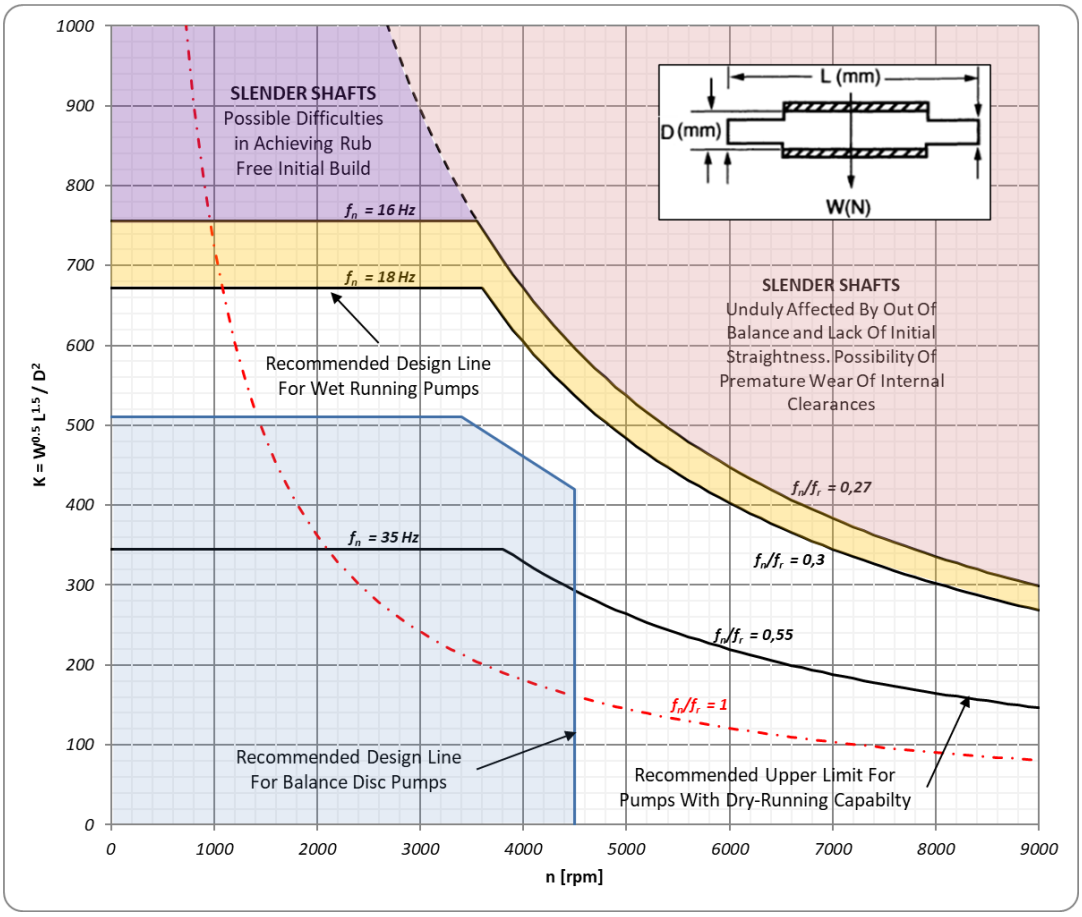

Figure 5: Duncan & Hood diagram.

A notable advantage of this factor is that some guidelines from Duncan and Hood works for limiting values have been defined. *Figure 5* allows a rotor to be classified as "too slender", "slender", suitable for "wet operation only," or capable of "running dry". Moreover, a recommended operating zone for balance disc pumps can be defined.

6. References

- *The Application of Recent Pump Developments to the Needs of the Offshore Oil Industry (A.B. Duncan, J. F. Hood)*
- *Centrifugal Pumps (I. J. Karassic, T. McGuire)*
- *API Standard 610 12th edition, January 2021 – Annex K.*
- ***Mod.496** – Preliminary rotor sizing - API610 and Duncan & Hood*

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